

Name - Saikat Sheet
Batch - 25SUB5936_Linux System
Programming
ID - 68500

1. Explain in simple words why IPv6 is important ?

Ans:

IPv6 is the modern "address book" for the internet. It is important because the old system (**IPv4**) ran out of unique addresses for all the world's devices.

Why it matters:

- **Infinite Space:** It provides trillions of new addresses so every **Smartphone**, **Laptop**, and **Smart TV** can have its own ID.
- **Faster Speed:** It helps your **Router** move data more efficiently.
- **Better Security:** It has safety features built-in to protect your data from hackers.

Simple Comparison:

- **IPv4:** Like a small town with no more house numbers available.
- **IPv6:** Like an infinite city where everyone can have a unique address.

2. What is the purpose of a subnet mask ?

Ans:

A **subnet mask** is a filter that separates an **IP address** into two parts: the **Network** (your street) and the **Host** (your specific device).

Why it matters:

- **Organization:** It tells your **Router** which devices belong to your home "neighborhood."
- **Traffic Control:** It helps your **Laptop** decide if data should stay inside your house or be sent out to the internet.
- **Security:** It creates a "border" that keeps your internal devices grouped together.

Simple Analogy:

- **IP Address:** Your full mailing address.

- **Subnet Mask:** The part of the address that identifies your **Zip Code**.

3. How does NAT help in home networks ?

Ans:

NAT (Network Address Translation) is a feature inside your **Wi-Fi Router** that allows all your devices to share a single internet connection.

Why it is important:

- **One Public Address:** Your Internet Provider only gives you **one** public IP address. NAT acts like a "mail clerk," allowing your **Laptop**, **Smart TV**, and **Mobile Phones** to all use that one address at the same time.
- **Private ID Numbers:** NAT gives each device in your house a "private" address (like 192.168.1.5) that only works inside your home.
- **Basic Security:** It hides your devices from the open internet. Since outsiders only see your router's address, they can't easily "see" or connect directly to your phone or laptop.

Simple Analogy:

- **The Internet:** A massive city.
- **Your Router (NAT):** The lobby of an apartment building.
- **Your Devices:** The individual apartments.
- **The Process:** All mail comes to the building's one street address (Public IP), and the lobby clerk (NAT) delivers it to the correct apartment number (Private IP).

4. When would you prefer powerline networking ?

Ans:

Best Situations for Powerline:

- **Thick Walls:** If your **Wi-Fi** signal cannot reach a back room because of brick or concrete walls, powerline carries the signal through the copper wires instead.
- **Gaming or Streaming:** If your **Smart TV** or gaming console is too far from the **Router** and experiences "lag," a powerline adapter provides a more consistent connection than Wi-Fi.
- **Desktop PCs:** If you have a computer in a home office without a network jack, you can simply plug an adapter into the wall outlet to get "wired" internet.

How it works:

1. Plug one adapter into a wall outlet near your **Router** and connect them with a short cable.
2. Plug the second adapter into an outlet near your **Laptop** or **Smart TV**.
3. The internet signal travels through the electrical wires in your walls to the other room

5. Why is fiber preferred for long-distance links?

Ans:

Fiber optic cables are preferred for long distances because they transmit data as **light** through glass, rather than electricity through copper.

Why it is better:

- **Distance:** Light signals can travel over 60 miles without fading; copper signals weaken after only a few hundred feet.
- **No Interference:** Glass does not pick up electrical noise from power lines or lightning, keeping the data clean.
- **Massive Speed:** Fiber can carry much more data at once, which is why it is used to connect entire cities and continents.

Simple Analogy:

- **Copper:** Like shouting through a tube; eventually, the sound becomes a whisper.
- **Fiber:** Like using a high-powered laser pointer; the beam stays bright and sharp for miles.